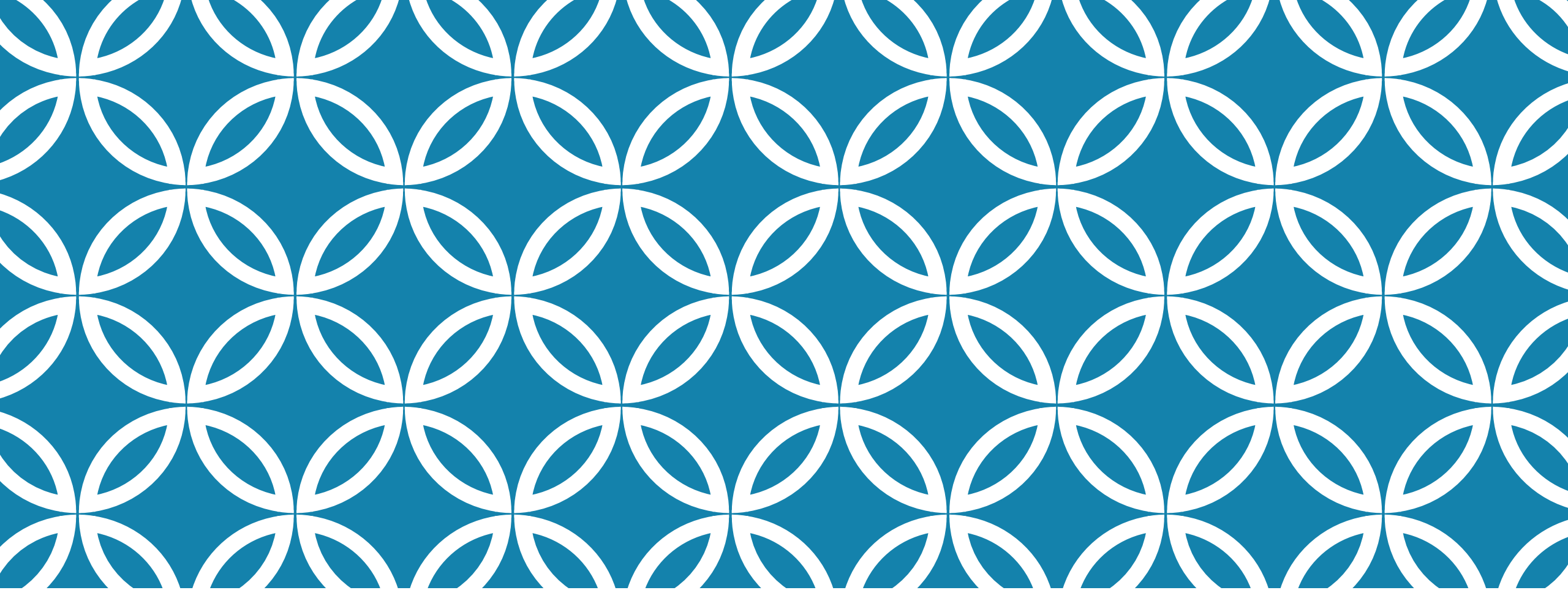


SEMANTIC CACHING FOR AI AGENTS

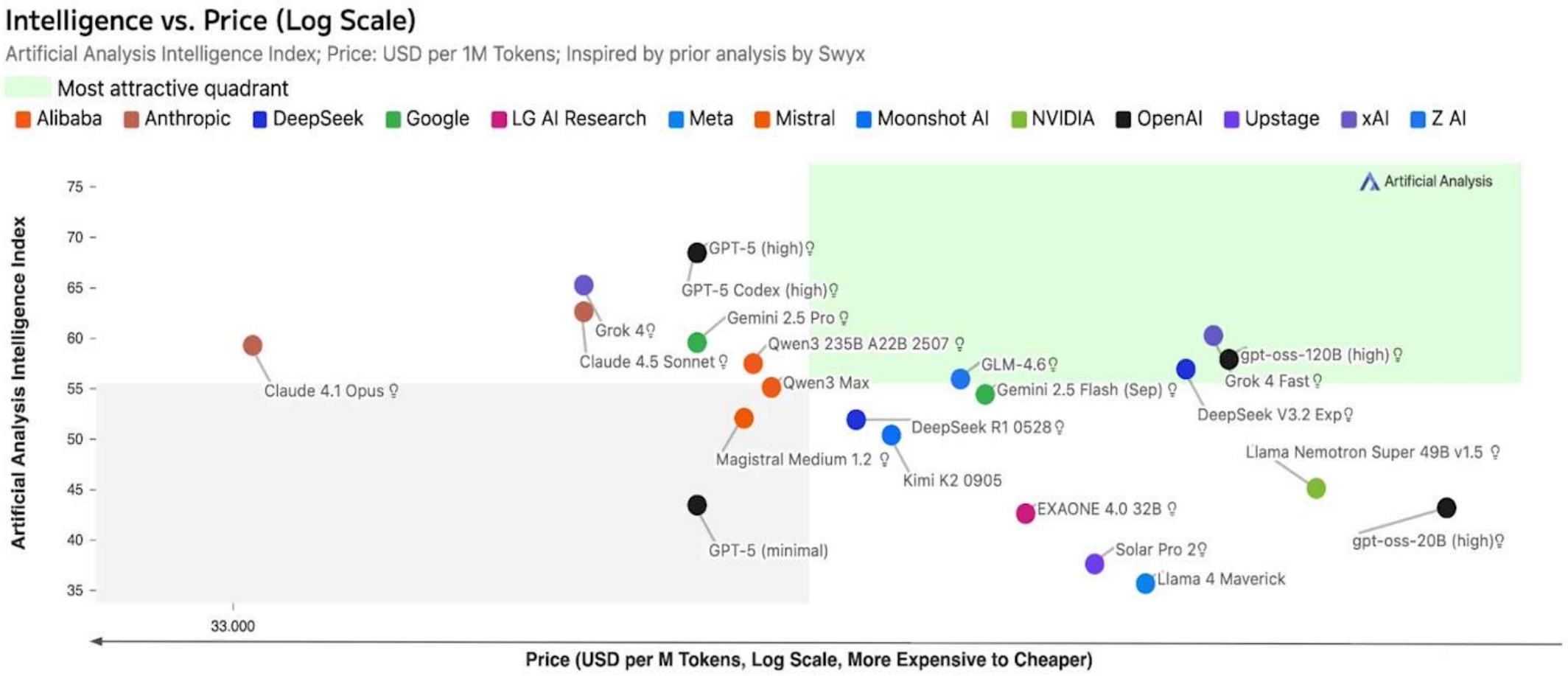
Fran Kufrin

23.12.2025.

PREGLED

- Motivacija
- Semantička priručna memorija
- Implementacija jednostavne semantičke priručne memorije
- Evaluacija semantičke priručne memorije
- Poboljšanje semantičke priručne memorije
- AI agent sa semantičkom priručnom memorijom

MOTIVACIJA

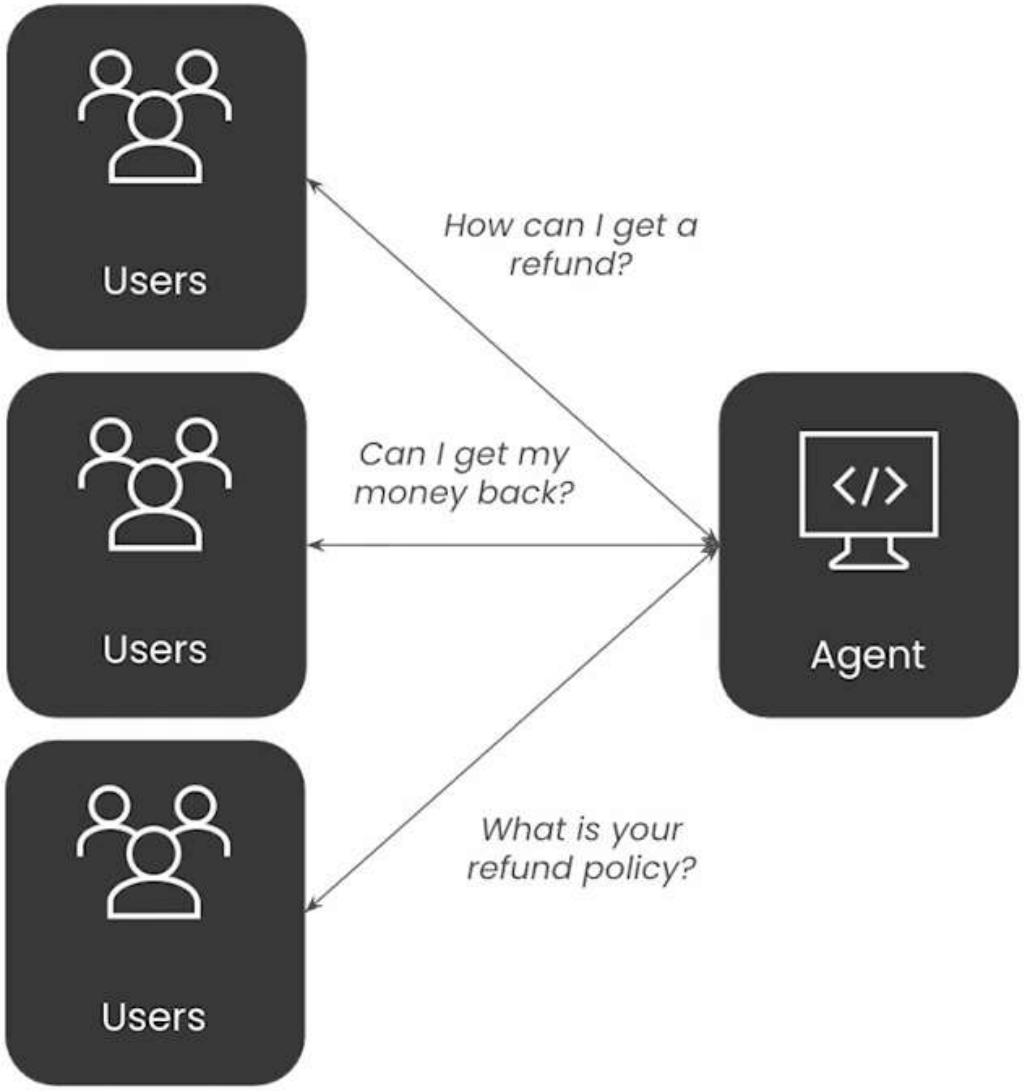

Intelligence vs. Price (Log Scale)
Artificial Analysis Intelligence Index; Price: USD per 1M Tokens; Inspired by prior analysis by Swyx
Most attractive quadrant
Alibaba
Anthropic
DeepSeek
Google
LG AI Research
Meta
Mistral
Moonshot AI
NVIDIA
OpenAI
Upstage
xAI
Z AI
Artificial Analysis
Artificial Analysis Intelligence Index
75
70
65
60
55
50
45
40
35
33.000
GPT-5 (high)
GPT-5 Codex (high)
Grok 4
Gemini 2.5 Pro
Claude 4.5 Sonnet
Qwen3 235B A22B 2507
Claude 4.1 Opus
Qwen3 Max
GLM-4.6
Gemini 2.5 Flash (Sep)
gpt-oss-120B (high)
Grok 4 Fast
DeepSeek V3.2 Exp
DeepSeek R1 0528
Magistral Medium 1.2
Kimi K2 0905
Llama Nemotron Super 49B v1.5
EXAONE 4.0 32B
GPT-5 (minimal)
Solar Pro 2
gpt-oss-20B (high)
Llama 4 Maverick
Price (USD per M Tokens, Log Scale, More Expensive to Cheaper)

MOTIVACIJA

- AI agenti koriste puno LLM poziva za generiranje rezultata
- Generiraju puno tokena
- Dodaju kašnjenje u generiraju odgovora korisniku
- Cijela izvedba od početka do kraja može koštati i do 6,8 USD.

PRIMJER: KORISNIČKA PODRŠKA

- Za svaki upit se poziva AI agent
- Može biti sporo i skupo
- Za slične upite se poziva AI agent umjesto da se iskoristi isti odgovor

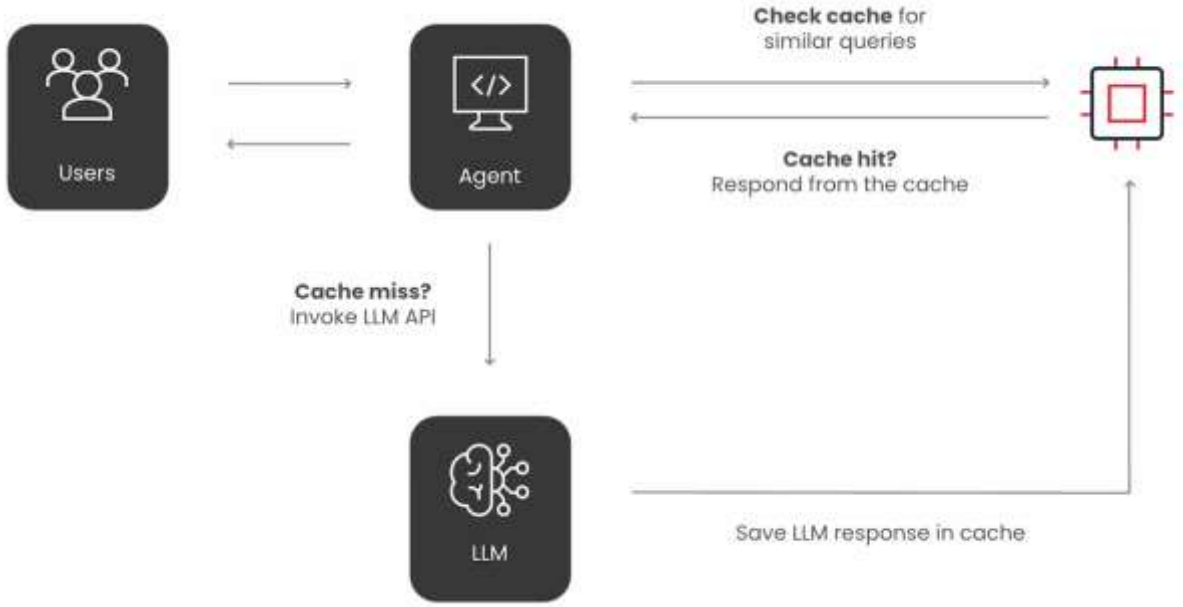


PRIRUČNA MEMORIJA

- Za iste upite se koristi odgovor spremljen u priručnoj memoriji

- Perfektna preciznost, niski odziv

- Problem: upiti moraju biti egzaktni za upotrebu priručne memorije -> mali CHR (Cache hit rate)



SEMANTIČKA PRIRUČNA MEMORIJA

- Pretraživanje priručne memorije na temelju semantičkog značenja upita
- Veći CHR (Cache hit rate)
- Uvodi mogućnost lažno pozitivnih rezultata (odgovor na upit nije traženi odgovor)

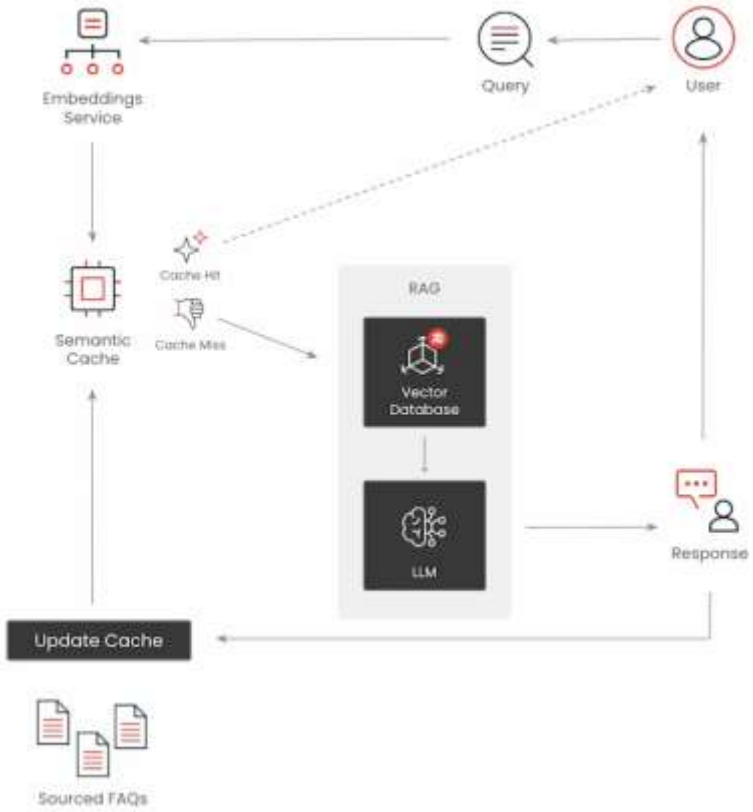
SEMANTIČKA PRIRUČNA MEMORIJA

1. Korisnikov upit pretvorimo u vektorsku reprezentaciju

2. Uspoređujemo vektor sa vektorskim reprezentacijama upita pohranjenih u priručnoj memoriji

3. Ako je ulazni upit dovoljno sličan nekom upitu u priručnoj memoriji onda vraćamo rezultat. Ako nije onda šaljemo upit LLM-u

4. Ažuriramo priručnu memoriju



IMPLEMENTACIJA JEDNOSTAVNE SEMANTIČKE PRIRUČNE MEMORIJE

```python
import pandas as pd
import numpy as np
import time

from cache.faq_data_container import FAQDataContainer

faq_data = FAQDataContainer()
faq_df = faq_data.faq_df
test_df = faq_data.test_df
```

```
Loaded 8 FAQ entries
Loaded 80 test queries
```

```python
faq_df.head().style
```

	id	question	answer
0	0	How do I get a refund?	To request a refund, visit your orders page and select **Request Refund**. Refunds are processed within 3-5 business days.
1	1	Can I reset my password?	Click **Forgot Password** on the login page and follow the email instructions. Check your spam folder if you don't see the email.
2	2	Where is my order?	Use the tracking link sent to your email after shipping. Orders typically arrive within 2-7 business days depending on your location.
3	3	How long is the warranty?	All electronic products include a 12-month warranty from the purchase date. Extended warranties are available for purchase.
4	4	Do you ship internationally?	Yes, we ship to over 50 countries worldwide. International shipping fees and delivery times vary by destination.

PODACI

UGRAĐENE REPREZENTACIJE UPITA

```python
from sentence_transformers import SentenceTransformer

encoder = SentenceTransformer("all-mpnet-base-v2")

faq_embeddings = encoder.encode(faq_df["question"].tolist())

print(f"Sample (first 10 dimensions): {faq_embeddings[0][:10]}")
```

```
Sample (first 10 dimensions): [ 0.02746389  0.04248321 -0.02679515  0.03786
725 -0.03876124 -0.00109459
 -0.01158362  0.03070287 -0.00653773 -0.01955771]
```

```python
def cosine_dist(a: np.array, b: np.array):
    """Compute cosine distance between two sets of vectors."""
    a_norm = np.linalg.norm(a, axis=1)
    b_norm = np.linalg.norm(b) if b.ndim == 1 else np.linalg.norm(b, axis=1
    sim = np.dot(a, b) / (a_norm * b_norm)
    return 1 - sim


def semantic_search(query: str) -> tuple:
    """Find the most similar FAQ question to the query."""
    query_embedding = encoder.encode([query])[0]

    distances = cosine_dist(faq_embeddings, query_embedding)

    # Find the most similar question (lowest distance)
    best_idx = int(np.argmin(distances))
    best_distance = distances[best_idx]

    return best_idx, best_distance
```

SEMANTIČKO PRETRAŽIVANJE

```
idx, distance = semantic_search(
    "How long will it take to get a refund for my order?"
)

print(f"Most similar FAQ: {faq_df.iloc[idx]['question']}")
print(f"Answer: {faq_df.iloc[idx]['answer']}")
print(f"Cosine distance: {distance:.3f}")
```




```
Most similar FAQ: How do I get a refund?
Answer: To request a refund, visit your orders page and select **Request Re
fund**. Refunds are processed within 3-5 business days.
Cosine distance: 0.331
```

SEMANTIČKO PRETRAŽIVANJE

```python
def check_cache(query: str, distance_threshold: float = 0.3):
    """
    Semantic cache lookup for previously asked questions.
    Returns a dictionary with answer if hit, None if miss.
    """
    idx, distance = semantic_search(query)

    if distance <= distance_threshold:
        return {
            "prompt": faq_df.iloc[idx]["question"],
            "response": faq_df.iloc[idx]["answer"],
            "vector_distance": float(distance),
        }

    return None  # Cache miss
```

PRIRUČNA MEMORIJA

```python
test_queries = [
    "Is it possible to get a refund?",
    "I want my money back",
    "What are your business hours?",  # Should miss
]

for query in test_queries:
    result = check_cache(query, distance_threshold=0.3)
    if result:
        print(f"✅ HIT: '{query}' -> {result['response'][:50]}...")
        print(f"   Distance: {result['vector_distance']:.3f}\n")
    else:
        print(f"❌ MISS: '{query}'\n")
```

```
Batches: 100% ████████████████████ 1/1 [00:00<00:00, 22.33it/s]
✅ HIT: 'Is it possible to get a refund?' -> To request a refund, visit you
r orders page and se...
   Distance: 0.262

Batches: 100% ████████████████████ 1/1 [00:00<00:00, 25.93it/s]
❌ MISS: 'I want my money back'

Batches: 100% ████████████████████ 1/1 [00:00<00:00, 22.77it/s]
❌ MISS: 'What are your business hours?'
```

PRIRUČNA MEMORIJA

```python
def add_to_cache(question: str, answer: str):
    """
    Add a new Q&A pair to our simple in-memory cache.
    Extends both the DataFrame and embeddings matrix.
    """
    global faq_df, faq_embeddings

    new_row = pd.DataFrame({"question": [question], "answer": [answer]})
    faq_df = pd.concat([faq_df, new_row], ignore_index=True)

    # Generate embedding for the new question
    new_embedding = encoder.encode([question])

    # Add to embeddings matrix
    faq_embeddings = np.vstack([faq_embeddings, new_embedding])

    print(f"✅ Added to cache: '{question}'")
```

DODAVANJE UPITA U PRIRUČNU MEMORIJU

```python
import os

REDIS_URL = os.getenv("REDIS_URL", "redis://localhost:6379")
```

```python
from redisvl.utils.vectorize import HFTextVectorizer
from redisvl.extensions.cache.embeddings import EmbeddingsCache

langcache_embed = HFTextVectorizer(
    model="redis/langcache-embed-v1",
    cache=EmbeddingsCache(redis_client=r, ttl=3600)
)
```

```python
MODEL_NAME = "gpt-4o-mini"

llm = ChatOpenAI(
    model=MODEL_NAME,
    temperature=0.1,
    max_tokens=150,
)
```

```python
from redisvl.extensions.cache.llm import SemanticCache

cache = SemanticCache(
    name="faq-cache",
    vectorizer=langcache_embed,
    redis_client=r,
    distance_threshold=0.3
)
```

```python
for i in range(len(faq_df)):
    cache.store(
        prompt=faq_df.iloc[i]["question"],
        response=faq_df.iloc[i]["answer"]
    )
```

```python
def get_llm_response(question: str) -> str:
    prompt = f"""
    You are a helpful customer support assistant. Answer this customer question concisely and professionally:

    Question: {question}

    Provide a helpful response in 1-2 sentences. If you don't have specific information, give a general helpful response.
    """
    response = llm.invoke([HumanMessage(content=prompt)])
    return response.content.strip()
```

```python
from cache.evals import PerfEval

perf_eval = PerfEval()

test_questions = [
    "How can I get my money back?",
    "I want a refund please",
    "What's your return policy?",
    "I forgot my password",
    "Can you help me reset my password?",
    "What are your shipping costs?",
    "Do you offer installation services?",
    "Can I schedule a phone call with support?",
    "How do I cancel my subscription?",
    "How much does shipping cost?",
    "I need to cancel my account",
]

perf_eval.set_total_queries(len(test_questions))
```

```python
with perf_eval:
    for i, question in enumerate(test_questions, 1):
        print(f"\n[{i}] Question: '{question}'")

        perf_eval.start()

        if cached_result := cache.check(question):
            # Cache HIT
            perf_eval.tick("cache_hit")
            print(
                f"    ✅ CACHE HIT (distance: {cached_result[0]['vector_distance']:.3f})"
            )
            print(f"    📋 Cached question: {cached_result[0]['prompt'][:80]}...")
            print(f"    📋 Cached response: {cached_result[0]['response'][:80]}...")
        else:
            # Cache MISS - call LLM
            perf_eval.tick("cache_miss")  # Time for cache check
            print(f"    ❌ CACHE MISS")
            print(f"    🤖 Calling LLM... ", end="")

            # Call LLM and track the call
            perf_eval.start()
            llm_response = get_llm_response(question)
            perf_eval.tick("llm_call")
            perf_eval.record_llm_call(MODEL_NAME, question, llm_response)
            print(f"    💬 LLM response: {llm_response[:80]}...")
            cache.store(prompt=question, response=llm_response)
```

```python
np.mean(perf_eval.durations_by_label['cache_hit'])
```

```
np.float64(0.06460990224565778)
```

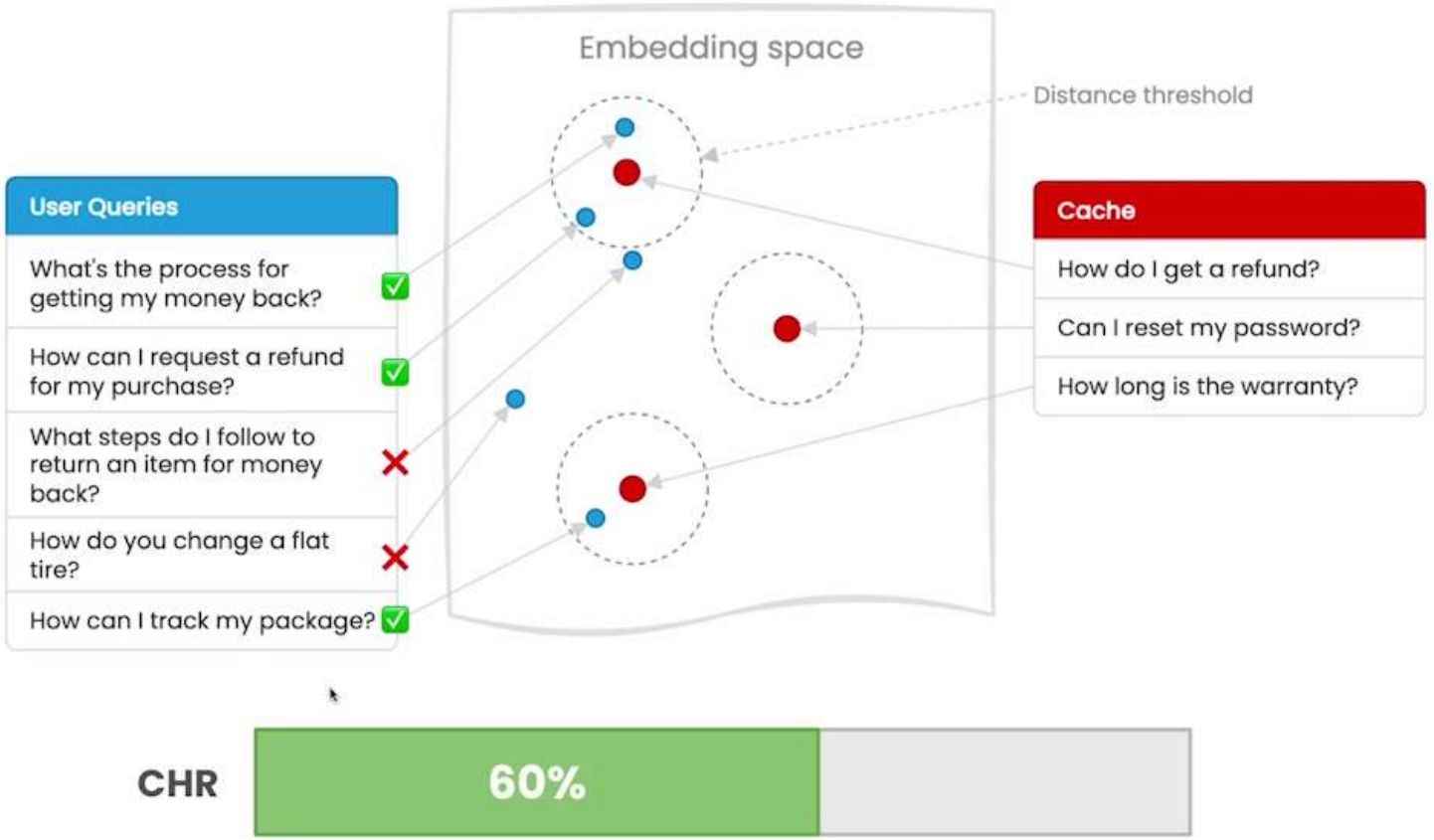
```python
np.mean(perf_eval.durations_by_label['llm_call'])
```

```
np.float64(1.3369968533515 93)
```

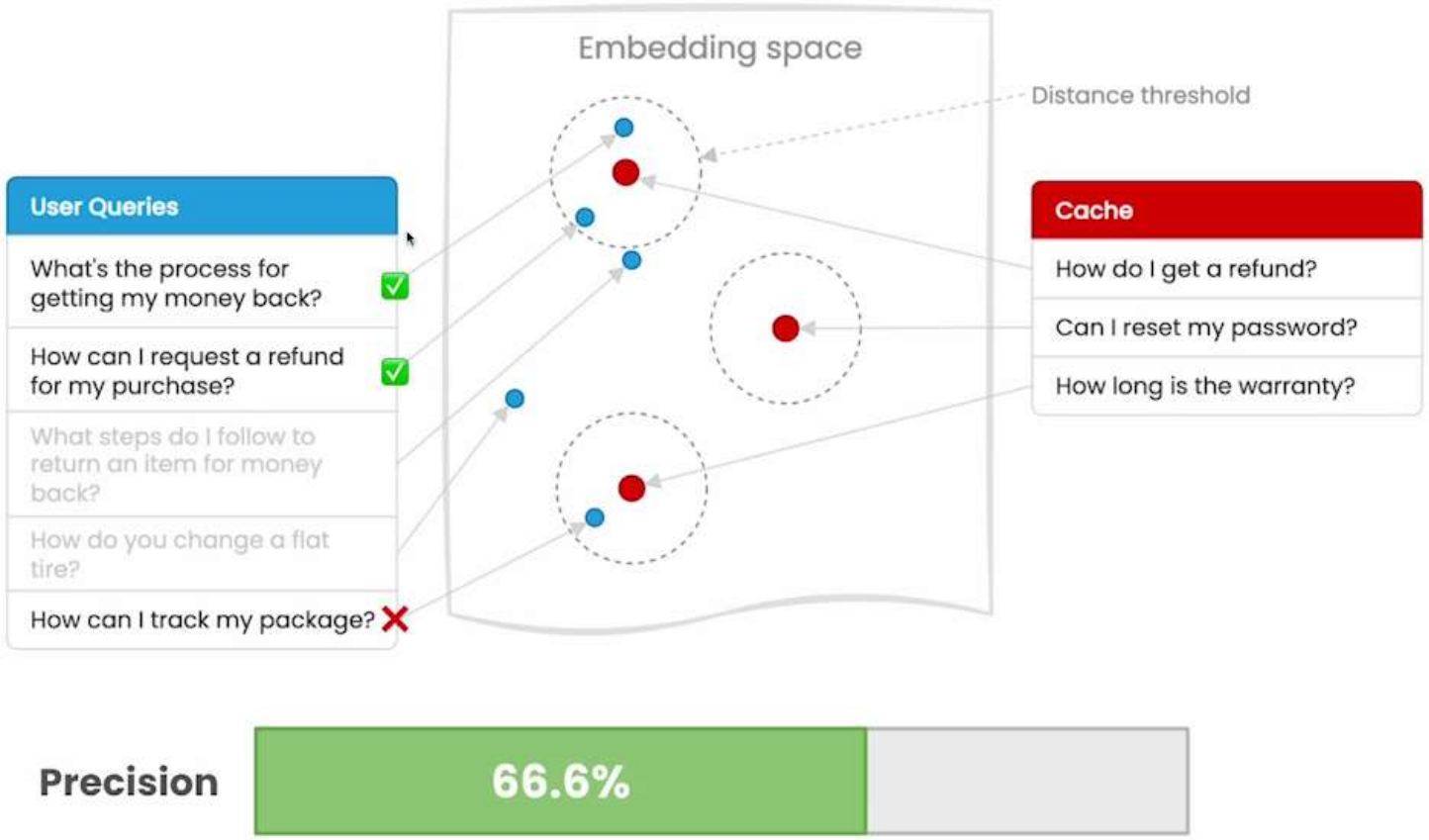
EVALUACIJA SEMANTIČKE PRIRUČNE MEMORIJE

PODACI ZA EVALUACIJU

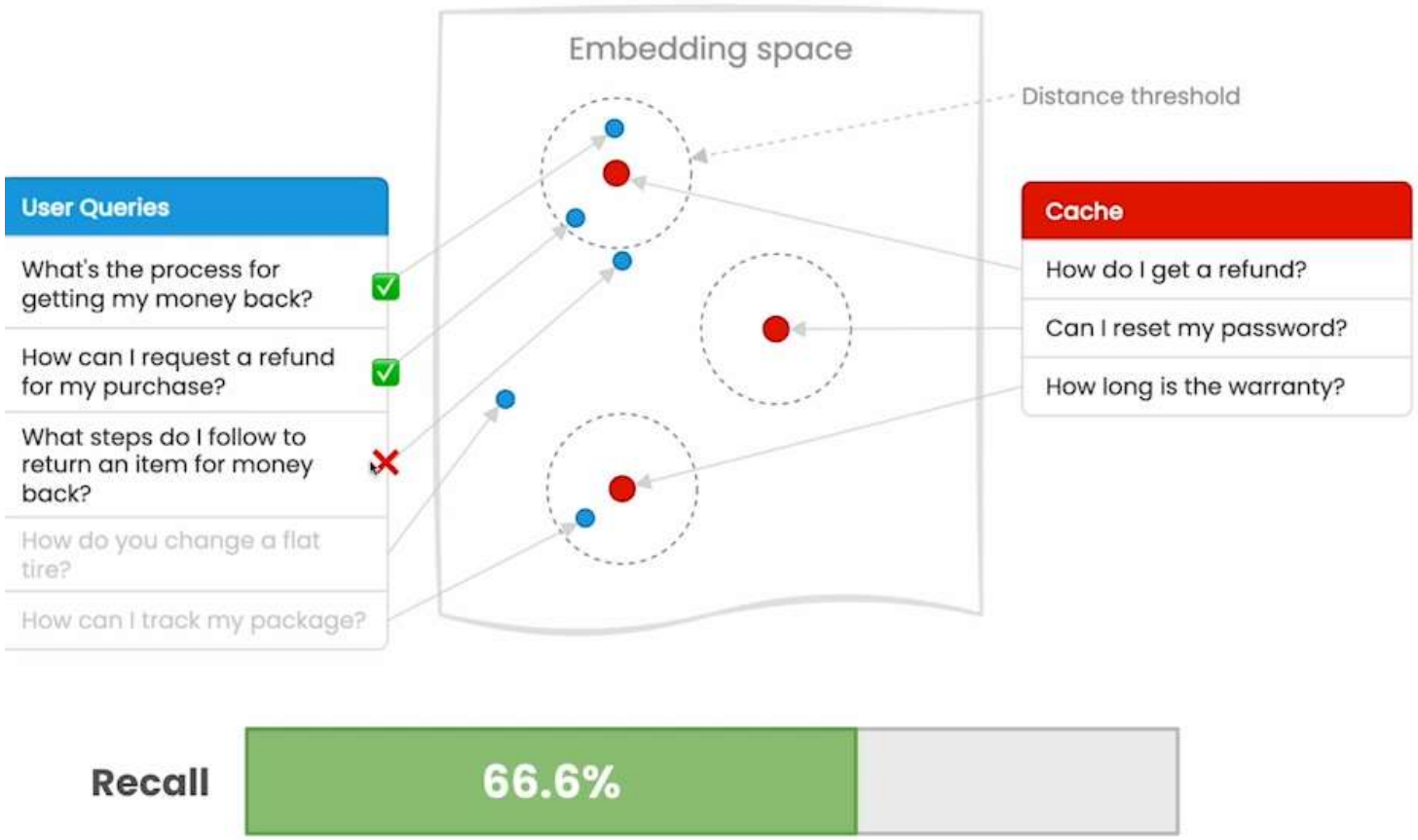
Query	Cache Hit	Distance	Label
What's the process for getting my money back?	How do I get a refund?	0.1	True
How can I request a refund for my purchase?	How do I get a refund?	0.2	True
What steps do I follow to return an item for money back?	How do I get a refund?	0.35	True
How do you change a flat tire?	How long is the warranty?	0.8	False
How can I track my package?	How long is the warranty?	0.15	False



CACHE HIT RATIO

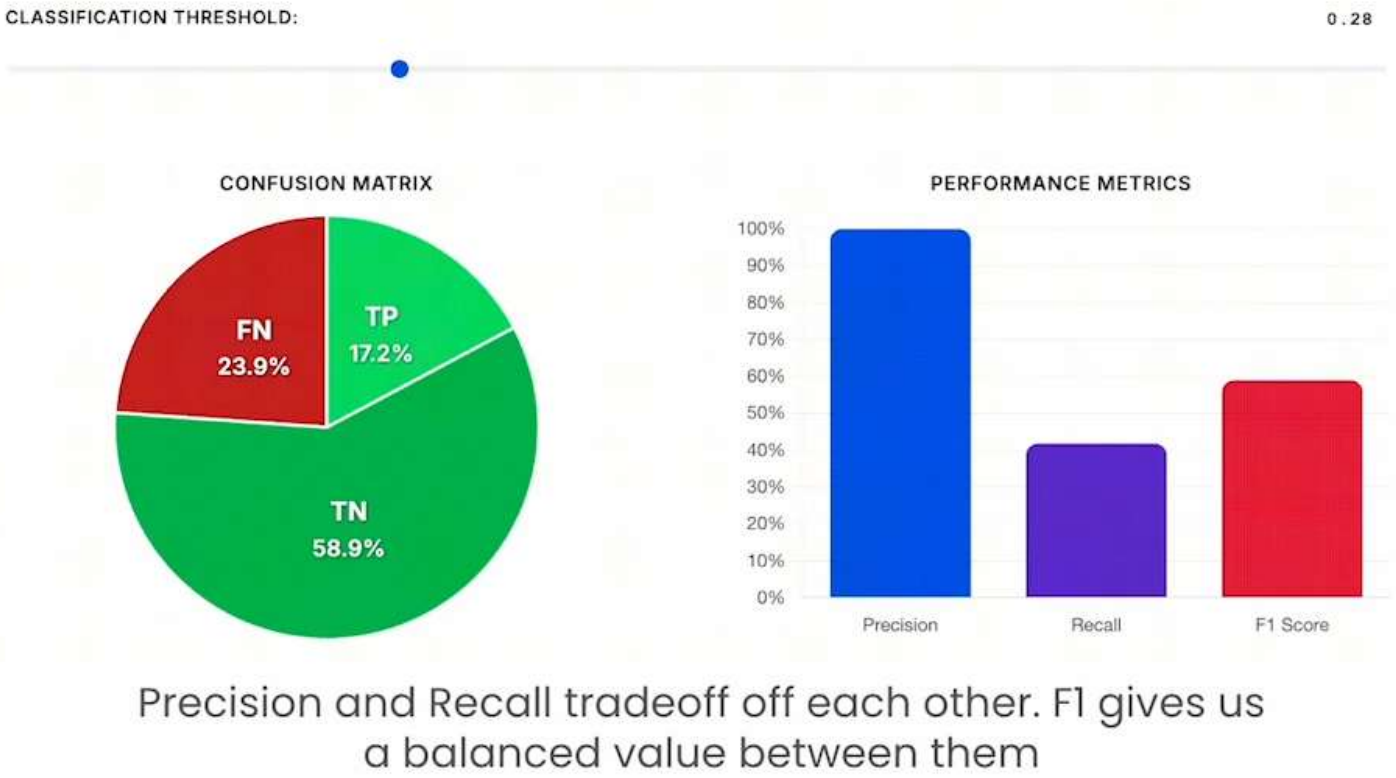

Embedding space
Distance threshold
User Queries
What's the process for getting my money back?
How can I request a refund for my purchase?
What steps do I follow to return an item for money back?
How do you change a flat tire?
How can I track my package?
Cache
How do I get a refund?
Can I reset my password?
How long is the warranty?
Precision
66.6%

PRECIZNOST


Embedding space
Distance threshold
User Queries
What's the process for getting my money back?
How can I request a refund for my purchase?
What steps do I follow to return an item for money back?
How do you change a flat tire?
How can I track my package?
Cache
How do I get a refund?
Can I reset my password?
How long is the warranty?

Recall
66.6%

ODZIV


CLASSIFICATION THRESHOLD:
0.28
CONFUSION MATRIX
FN
23.9%
TP
17.2%
TN
58.9%
PERFORMANCE METRICS
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
Precision
Recall
F1 Score
Precision and Recall tradeoff off each other. F1 gives us
a balanced value between them

F1-MJERA

LAŽNO POZITIVNI REZULTATI

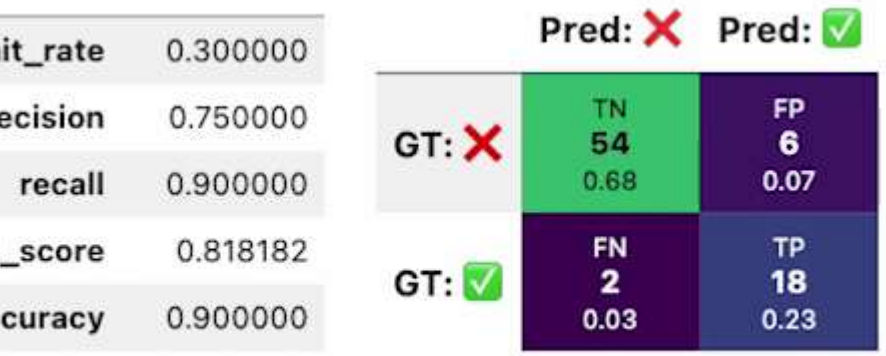
	query	match	distance	true_label
5	Can I get a refund if I changed my mind?	How do I get a refund?	0.294378	False
24	Can I change my order after it's been placed?	Can I change my delivery address?	0.286985	False
53	How do I upgrade my subscription plan?	How do I cancel my subscription?	0.250155	False
55	Can I pause my subscription temporarily?	How do I cancel my subscription?	0.277127	False
75	Can I schedule a specific delivery time?	Can I change my delivery address?	0.286493	False

LLM-AS-A-JUDGE

- LLM se koristi za završnu odluku da li će se vratiti rezultat iz priručne memorije

- Vraća odgovor da/ne na pitanje da li vektori imaju isto semantičko značenje

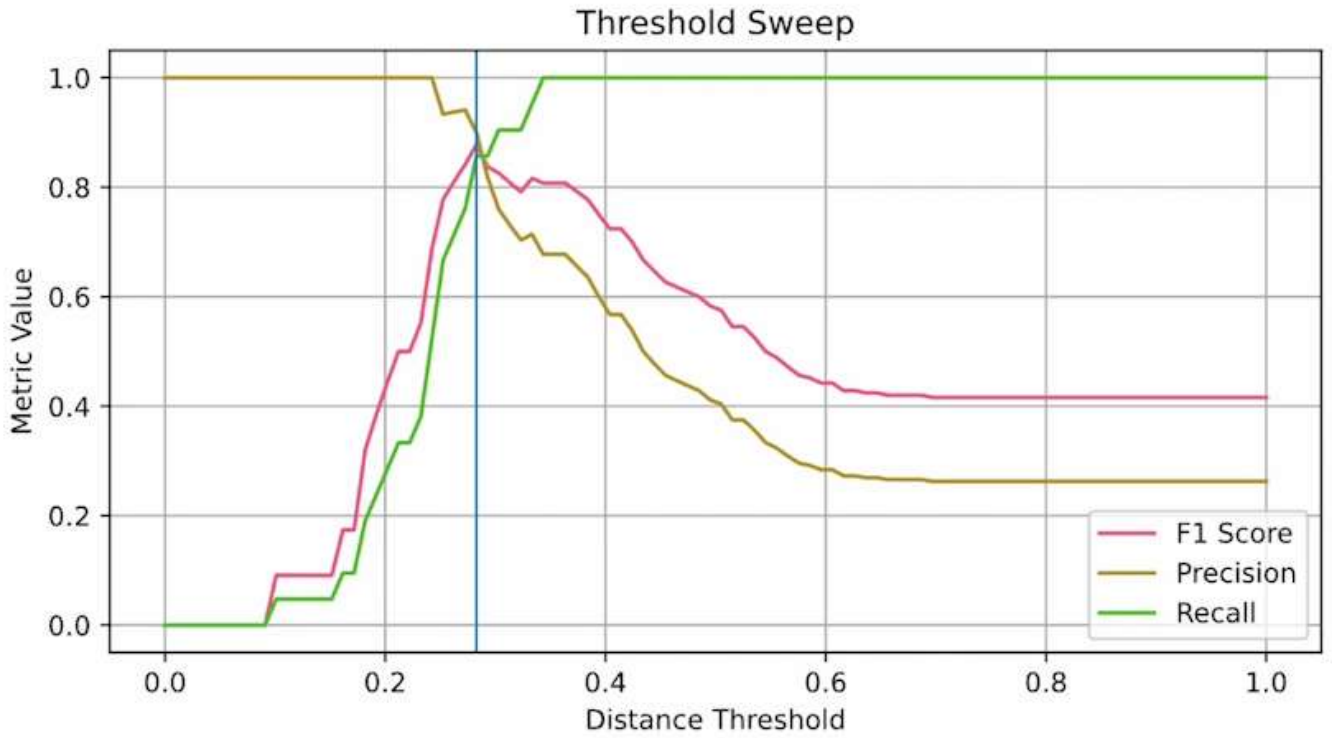
cache_hit_rate	0.300000
precision	0.750000
recall	0.900000
f1_score	0.818182
accuracy	0.900000
utility	0.428571



NAČINI ZA POBOLJŠANJE PERFORMANSI

PRETRAŽIVANJE NAJBOLJEG PRAGA

REDOM PRETRAŽUJEMO PRAG KOJI OSTVARUJE NAJBOLJU F1 MJERU



PONOVNO RANGIRANJE POMOĆU CROSS-ENKODERA

- Cross-enkoder daje bolje odluke sličnosti između dva upita

- Nekoliko najsličnijih primjera iz priručne memorije koje smo odredili preko vektorskih reprezentacija ponovno rangiramo pomoću cross-enkodera

Query	Cache hits	Distance	Cross-Encoder Score
What's the process for getting my money back?	How do I get a refund?	0.2	0.96
	Can I reset my password?	0.4	0.1
	How long is the warranty?	0.73	0.01

NEIZRAZITO PODUDARANJE (FUZZY MATCHING)

-Provodi se prije vektorizacije upita

-Leksička (tekstualna) sličnost, bez razumijevanja značenja

-Hvata tipfelere i near-duplicate upite

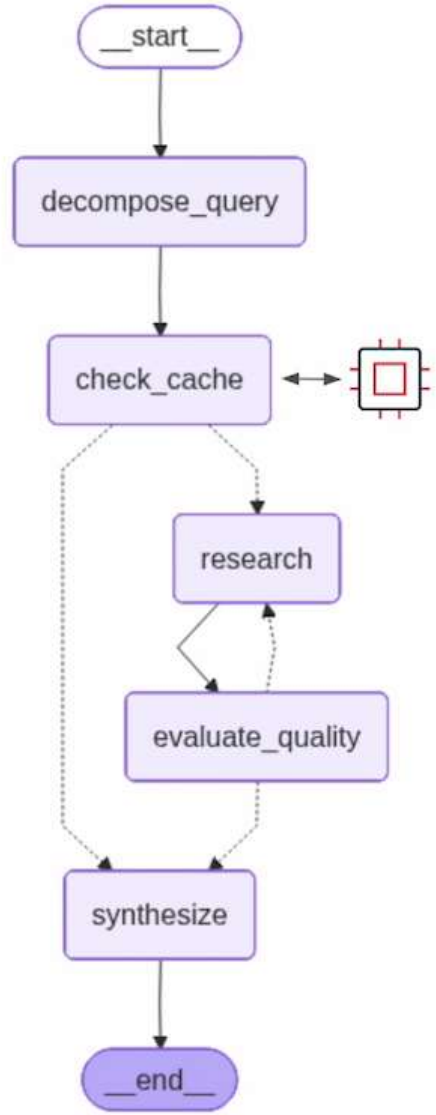


AI AGENT SA SEMANTIČKOM PRIRUČNOM MEMORIJOM

1. Rastavljamo veliki ulazni upit u 3-5 manjih podupita

2. Za svaki podupit provjerimo priručnu memoriju

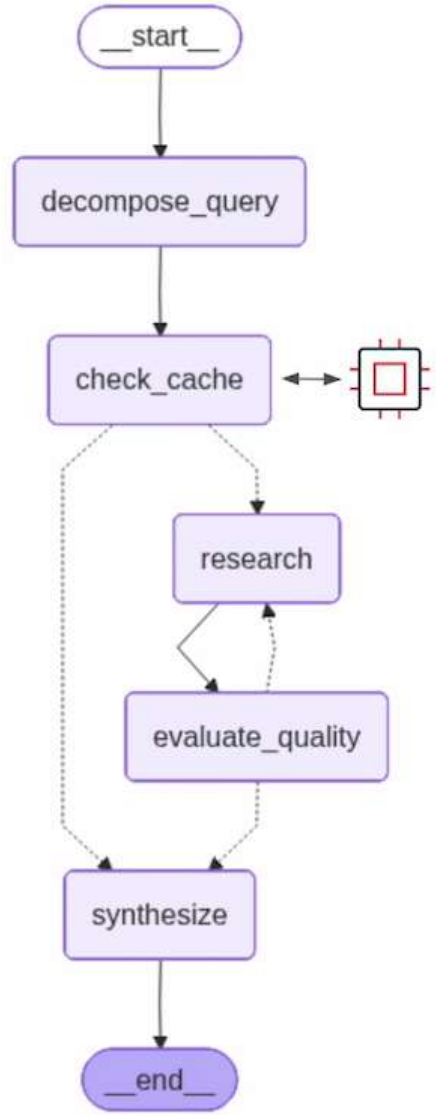
3. Podupite za koje nemamo odgovore u priručnoj memoriji šaljemo LLM-u



AI AGENT SA SEMANTIČKOM PRIRUČNOM MEMORIJOM

4. Evaluiramo odgovor iz LLM-a i koristimo LLM Judge za određivanje kvalitete odgovora. Ako kvaliteta odgovora nije dobra onda se ponovno šalje podupit LLM-u

5. LLM kombinira odgovore na podupite u jedan veliki odgovor



```python
scenario1_query = """
I'm evaluating your platform for our enterprise needs. We need to understand your
security and compliance standards including SOC2 status, want to know the API rate
limits for the Pro plan since we're considering that tier, need details about
Salesforce integration capabilities, and want to understand your payment options
including whether you support ACH transfers.
"""

result1 = run_agent(workflow_app, scenario1_query)
display_results(result1)
```

```
\n🎯 SEMANTIC CACHING WORKFLOW ANALYSIS
================================================================================
=====
📝 **Original Query:**
I'm evaluating your platform for our enterprise needs. We need to understan
d your
security and com...
🔄 **Execution Path:** decomposed → cache_checked → researched → quality_ev
aluated → synthesized
\n🧠 **Query Decomposition:** 4 sub-questions
   1. 🔍 **RESEARCH**: What are your security and compliance standards, inc
luding your current SOC2 status?
   2. 🔍 **RESEARCH**: What are the API rate limits for the Pro plan?
   3. 🔍 **RESEARCH**: What are your Salesforce integration capabilities?
   4. ✅ **CACHE HIT** (0.888): What payment options do you offer, and do y
ou support ACH transfers?
\n📊 **Performance Metrics:**
   💾 Cache Hit Rate: **25.0%** (1/4 questions)
   🤖 LLM Calls: **8**(GPT-4: 2, GPT-4-Mini: 6)
   ⚡ Total Latency: **21365.16ms**
   ⏱️  Cache: 989ms, Research: 8896ms
\n📄 **AI Response:**
--------------------------------------------------------------------------------
-----
```

Thank you for considering our platform for your enterprise needs! I'm happy to provide a comprehensive overview addressing your questions about security, API rate limits, Salesforce integration, and payment options.

First, regarding security and compliance, our platform is SOC2 compliant and adheres to robust standards including end-to-end encryption, GDPR compliance, SSO integration, audit logs, and IP whitelisting. If you have any specific security or compliance requirements, please let us know—we're happy to provide more detailed documentation or address particular concerns.

For the Pro plan, the API rate limit is 10,000 requests per hour. This should provide ample capacity for most enterprise use cases, but if you anticipate higher volumes or have unique integration needs, we can discuss options to ensure seamless performance.

When it comes to Salesforce integration, we offer a range of capabilities. We can guide you through connecting Salesforce with other systems via APIs, assist with data synchronization, and advise on using tools like Salesforce Connect, MuleSoft, or third-party integration platforms. We also support workflow automation involving Salesforce data and can help with custom Salesforce applications or components. If you have a specific integration scenario in mind, please share more details so we can provide tailored guidance.

Finally, for payment options, we accept all major credit cards, PayPal, Apple Pay, Google Pay, and bank transfers (including ACH transfers) for orders over $100.

If you need further details on any of these topics or have additional questions, please let me know—I'm here to help!

IZVORI

[1] https://www.deeplearning.ai/short-courses/semantic-caching-for-ai-agents/

[2] https://www.catchpoint.com/blog/semantic-caching-what-we-measured-why-it-matters

HVALA NA POZORNOSTI